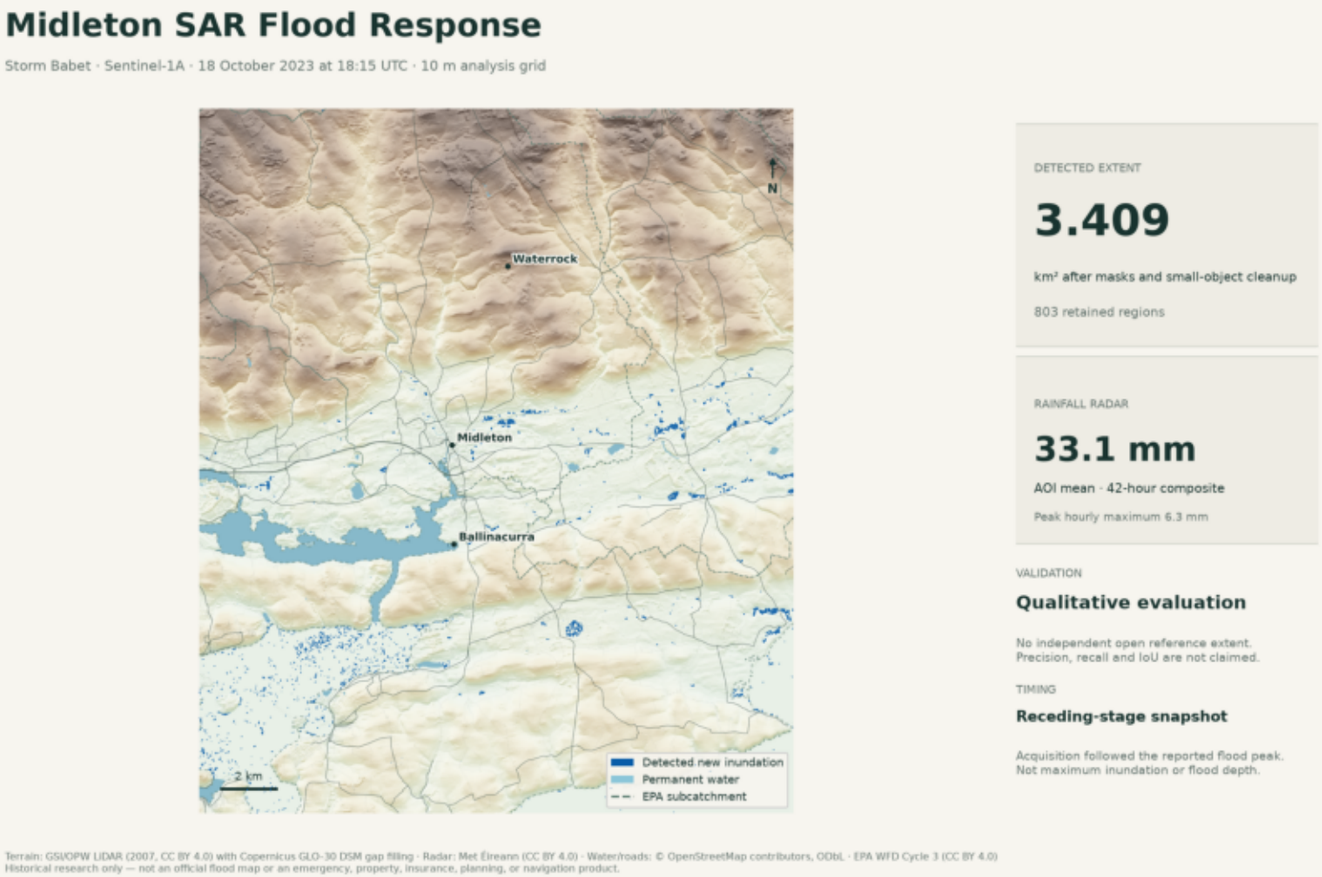
STORM BABET · MIDLETON

Rapid flood mapping with SAR + rainfall radar

Historical event response · 18 October 2023 · Ireland



Terrain: GSI/OPW LiDAR (2007, CC BY 4.0) with Copernicus GLO-30 DSM gap filling · Radar: Met Éireann (CC BY 4.0) · Water/roads: © OpenStreetMap contributors, ODbL · EPA WFD Cycle 3 (CC BY 4.0)
Historical research only — not an official flood map or an emergency, property, insurance, planning, or navigation product.

DETECTED NEW INUNDATION
3.409 km²

SAR ACQUISITION
18 Oct · 18:15 UTC

BASELINE
19 Aug · 24 Sep · 6 Oct

RADAR WINDOW
42 complete hours

AOI RADAR MEAN
33.1 mm

VALIDATION
Qualitative only

METHOD

SNAP orbit/noise correction → σ^0 calibration → Refined Lee speckle filtering → 10 m terrain correction → three-scene median baseline → VV water + negative-change thresholds → slope, permanent-water and layover/shadow exclusions → small-region cleanup → rainfall comparison and qualitative review

LIMITS

The satellite acquisition followed the reported flood peak; this is a receding-stage snapshot, not maximum inundation. No independent open reference extent was redistributed, so precision, recall and IoU are not claimed. Urban vegetation, buildings, rough water and radar geometry affect detectability. Not for emergency, property, insurance, planning or navigation use.

Data: Copernicus Sentinel-1 · Met Éireann radar (CC BY 4.0) · GSI/OPW DTM (CC BY 4.0) · EPA WFD Cycle 3 (CC BY 4.0) · © OpenStreetMap contributors (ODbL)